MATHEMATICS

The Honest Coder

Maths trick #7

Website → https://thehonestcoder-edu.github.io/

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Trick: Cube Root of numbers upto 6 digits

Step by Step Explanation

(with examples)

1. Divide the number into two parts - last 3 digits and remaining 1-3 digits at the beginning.

132|651

2. Find the highest perfect cube lesser than Group 1 (first 1-3 digits). Its cube root is the first digit.

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3. Find the digit whose cube ends in the units digit of Group 2 (last 3 digits). This digit will be the last digit of the cube root.

NOTE - If the procedure felt a bit complex, dont worry, the examples will help make sense.

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Units digit Power Map

1	1
2	8
3	7
4	4
5	5
6	6
7	3
8	2
9	9
0	0

Example 1→

$\sqrt[3]{804357}$

Step 1 - Make two groups -

- Group 1: first 3 digits (804)
- Group 2: last 3 digits (357)

Step 2 - The biggest perfect cube smaller than 804 is 729, which is the cube of 9.

Step 3 - The only digit whose cube ends with 7(the last digit of group 2) is 3. Therefore the second digit is 3.

Final answer → 93

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Example 2→

$\sqrt[3]{110592}$

Step 1 - Make two groups -

- Group 1: first 3 digits (110)
- Group 2: last 3 digits (592)

Step 2 - The biggest perfect cube smaller than 110 is 64, which is the cube of 4.

Step 3 - The only digit whose cube ends with 2(the last digit of group 2) is 8. Therefore the second digit is 8.

Final answer → 48

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Example 3→

$\sqrt[3]{512000}$

Step 1 - Make two groups -

- Group 1: first 3 digits (512)
- Group 2: last 3 digits (000)

Step 2 - The biggest perfect cube smaller than or equal to 512 is 512 itself, which is the cube of 8.

Step 3 - The only digit whose cube ends with 0(the last digit of group 2) is 0. Therefore the second digit is 0.

Final answer → 80

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Practice sheet

1. $\sqrt[3]{195112}$
2. $\sqrt[3]{405224}$
3. $\sqrt[3]{15625}$
4. $\sqrt[3]{778688}$
5. $\sqrt[3]{238328}$

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